

Human Development Index (HDI) Predictor

Technology Stack

Project: Human Development Index (HDI) Predictor
Phase: Requirement Analysis
Developer: Nainaru Thulasi Ram

1. Technology Stack Overview

The HDI Predictor uses a Python-based technology stack for data processing, machine learning, and web deployment.

2. Programming Languages

Language	Purpose
Python 3	Data processing, model training, and Flask web backend
HTML / CSS	Frontend structure and styling for the Home, Prediction, and Result pages
Jinja2 templating	Dynamic rendering of country dropdown and prediction results

3. Data Processing & ML Libraries

Library	Purpose
Pandas	Loading, cleaning, and manipulating the HDI dataset
NumPy	Numerical arrays and mathematical operations
Scikit-learn	LinearRegression, LabelEncoder, train_test_split, evaluation metrics
Pickle	Serializing/deserializing the trained model and country encoder

4. Visualization Libraries

Library	Purpose
Matplotlib	Actual-vs-predicted scatter plot, base plotting engine
Seaborn	Strip plot (schooling vs HDI) and correlation heatmap

5. Web Framework & Deployment

Component	Purpose
Flask	Lightweight Python web framework for routing and template rendering
Jinja2	Template engine (built into Flask) for dynamic HTML generation

6. Development Tools

Tool	Purpose
VS Code	Primary IDE for writing and running Python and HTML code
Git & GitHub	Version control and remote repository for the project